

Solution Explanation – The Somali Boys

Smallholder farmers in South Africa face a severe information gap. While market prices, rainfall, and fuel costs are publicly available, commercial operations hire analysts to interpret them, leaving smallholders reliant on buyer-dictated prices. Our solution in Kotulo, a web-based platform, bridges that gap by converting raw market data into one plain-language recommendation: SELL or HOLD. Requiring no statistical knowledge, setup, or login, a farmer simply selects their province, loads their data, and clicks a button to receive guidance in their preferred language—including English, Afrikaans, isiXhosa, isiZulu, or Sesotho.

Behind the scenes, Kotulo combines three forecasting models and automatically weights them by recent accuracy. It compares the blended price forecast against local storage costs and safety margins, recommending HOLD only when models overwhelmingly agree that waiting will yield a net profit after storage fees. Because an incorrect decision to hold costs real money in storage and price risk, the system conservatively defaults to SELL to prevent financial loss.

The architecture is highly scalable and commodity-neutral, making it quick and easy to expand beyond white maize to other sectors like livestock or citrus. Looking ahead, if Kotulo is ever monetized, that revenue could fund live data feeds directly from the Johannesburg Stock Exchange (JSE). Integrating live market data would completely eliminate the need for manual file uploads, turning the app into an effortless, zero-input tool for the farmer.

Ultimately, Kotulo distributes institutional-grade data analysis tools directly to smallholders. By delivering a clear, model-backed answer on whether to sell immediately or wait, the tool helps farmers avoid costly timing mistakes, protect their profit margins, and secure their livelihoods.